CHEMISTRY ADAPTLY

CHEMICAL ANALYSIS – FLAME TESTS & METAL IONS

KEY VOCABULARY

Flame test: the flame colour identifies a metal ion.

Lithium: crimson red flame.

Sodium / Potassium: yellow / lilac flame.

Hydroxide test: NaOH gives coloured precipitates with some ions.

COMMON MISCONCEPTIONS

~~Flame tests identify non-metal ions.~~

✓ They identify metal (positive) ions.

~~Two ions in a mix are easy to see by flame.~~

✓ One colour can mask another — a limitation.

~~All hydroxide precipitates are white.~~

✓ Copper is blue, iron(II) green, iron(III) brown.

ION TESTS

Li^+ crimson · Na^+ yellow · K^+ lilac

Ca^{2+} orange-red · Cu^{2+} green

With NaOH(aq): Cu^{2+} → blue precipitate.

Fe^{2+} → green · Fe^{3+} → brown precipitate.

Al^{3+}/Ca^{2+}/Mg^{2+} → white precipitate.

1 FLAME COLOUR

State the flame colour for **lithium** and **potassium**. **[2 marks]**

2 IDENTIFY THE ION

A flame test gives a yellow flame. **Identify** the metal ion. **[1 mark]**

3 HYDROXIDE TEST

State the precipitate colour when NaOH is added to Cu^{2+} and Fe^{3+}. **[2 marks]**

4 LIMITATION

Explain one limitation of using flame tests on a mixture. **[2 marks]**

5 FILL THE GAPS

Complete the sentence using the word bank. **[2 marks]**

With NaOH, iron(II) gives a _____ precipitate and iron(III) gives a _____ one.

blue | green | brown | white

6 WHY PRECIPITATE

Explain why an insoluble metal hydroxide forms when NaOH is added. **[1 mark]**

TIP: *Li crimson, Na yellow, K lilac, Ca orange-red, Cu green.*

✓ **Think: Why are instrumental methods more reliable than flame tests for a mixture?**